

Grown, not trained: what a developmental map makes likely before selection

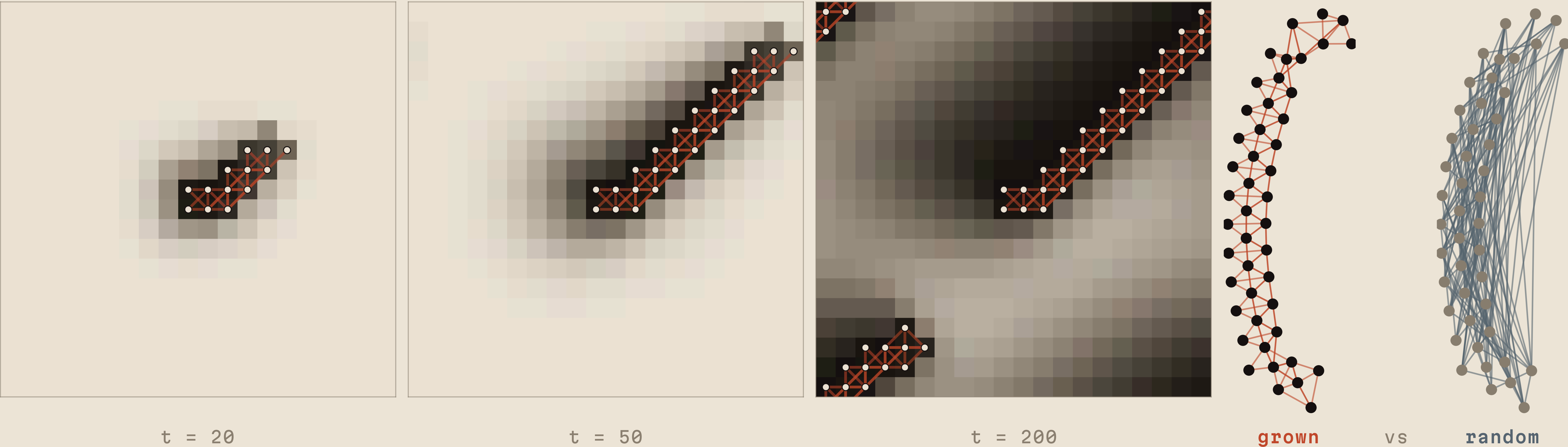
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Before any selection, this map raises its networks’ odds of success 17.6-fold over matched random wiring. Change what counts as success, and the advantage can flip sign.



Genomes do not specify synapses one by one; they specify developmental rules, and the genotype-to-phenotype map biases what selection gets to see. We measure that bias functionally: sample un-evolved genomes, grow them, and ask which outcomes the grown networks reach more often than random wiring of the same size.

WHAT GROWS

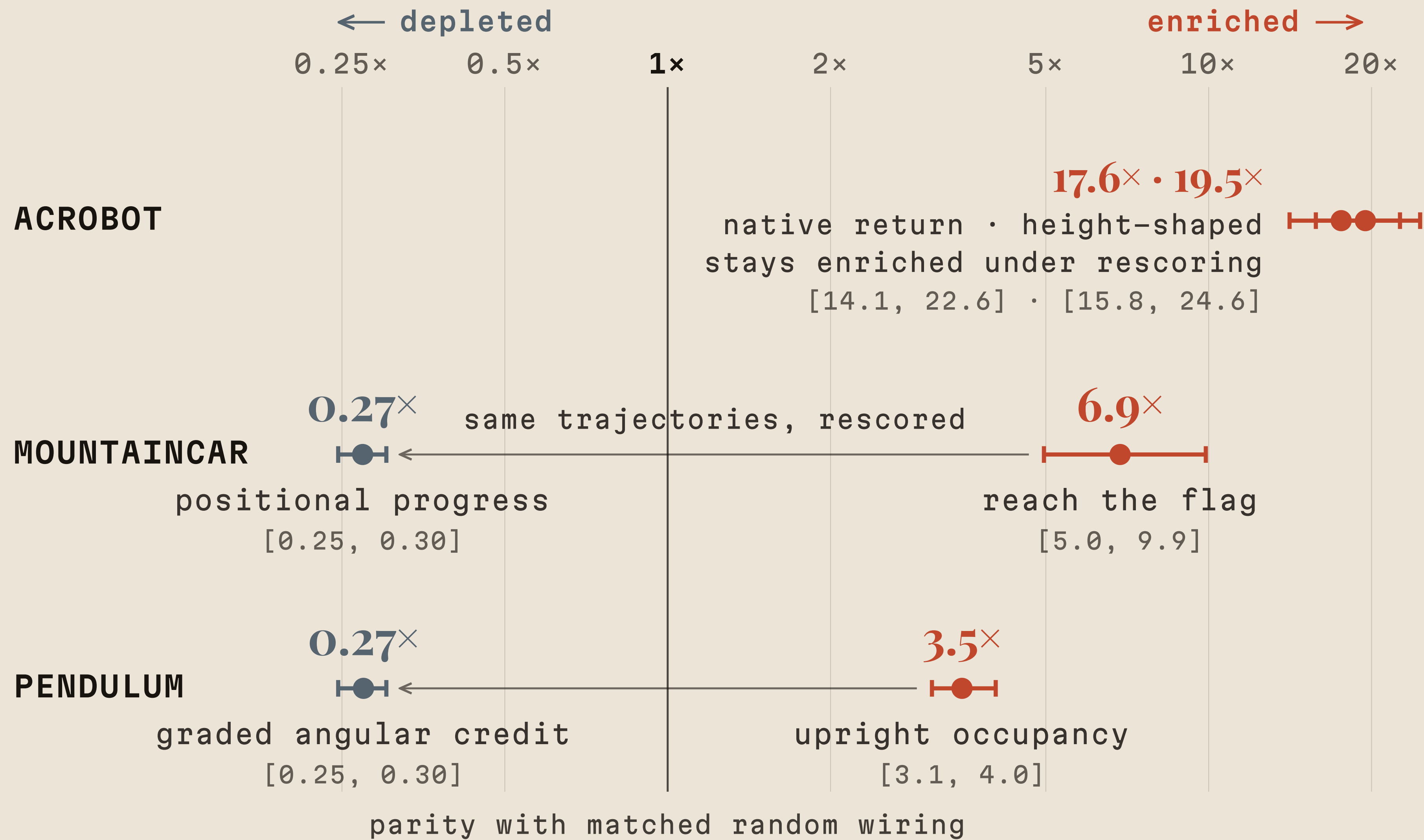


One un-evolved MorphoNAS genome: reaction-diffusion patterning lays down morphogen fields; neurons differentiate and wire along them. Right: the grown network beside a random digraph with the same neuron and edge count (N=46, E=155). Matched in size, unmatched in everything else. Illustrative, not evidence.

THE PRIOR EXISTS



CHANGE WHAT COUNTS AS SUCCESS



Same networks, same trajectories, rescored under a different definition of success (verified byte-identical). Structured targets: enriched. Dense progress credit: depleted. MountainCar 6.9× to 0.27×; Pendulum 3.5× to 0.27×; height-shaped Acrobot stays at 19.5×.

HOW WE MEASURED IT

- 5,000 genomes from the published initializer; canonical ruleset plus two evaluation fixes; no evolution, no learning, no plasticity.
- Recurrent rollout $s \leftarrow \phi(Ws)$, argmax over output nodes, 20 fixed seeds per network.
- Null: for every grown source, k=5 random digraphs matched on exact (N, E), uniform weights; draws with no weakly connected graph in 100 tries are dropped.
- Mantel-Haenszel odds ratio over exact (N, E) strata; 20,000-replicate source-cluster bootstrap CIs.
- The null controls size and density only: degree, reachability, locality, motifs, weights all unmatched.

LIMITS

- The null shares only size and density: the effect belongs to the whole initializer-plus-map pipeline, not morphogenesis in general, and no topological mechanism is isolated yet.
- Is it just structure? Partly: CPPN-patterned wiring also beats structureless random, and the grown map exceeds it (ongoing). Degree- and motif-preserving nulls are next.
- Five of six cells lack threshold sweeps; where score distributions cross, the sign depends on cut placement.

The map does not make networks good in general; it makes particular outcomes reachable.



Code + data · v0.1
github.com/ukma-morphonas-lab/MorphoNAS-DevPriors
Full paper in preparation · s.medvid@ukma.edu.ua